

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Python** scripts for data extraction and transformation, supporting automated generation of **3** key operational reports per week.
- Designed and implemented **Java Spring Boot** REST APIs to facilitate inter-service communication, handling up to **1,500** requests per minute during peak loads.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, ensuring adherence to SLAs and performing root cause analysis for P1/P2 issues.
- Implemented automated data ingestion and pre-processing pipelines using **Python** and **SQL**, preparing datasets for neural network training as instructed by the team.
- Executed training runs for neural networks with various tuning parameters on GPU instances, meticulously documenting experimental configurations and observations.
- Analyzed and summarized results from multiple training experiments, identifying optimal parameter sets and presenting findings to the R&D team.
- Containerized microservices using **Docker** and managed their deployment to **AWS EC2** instances, configuring basic resource limits for predictable performance.

SKILLS

Languages – Python, Java, C/C++, SQL, R

Frameworks/Libraries – Spring Boot, scikit-learn, pandas, NumPy, Flask, FastAPI, React, Angular

Cloud/Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux, Jenkins

Databases – PostgreSQL, MySQL, Oracle

ML/AI – Neural Networks, Generative Models, Photorealism, GPU Programming, OpenGL, Vulkan, Machine Learning

Tools/Methodologies – Git, Jira, Confluence, Agile, Scrum, CI/CD, OOP, Clean Coding Practices

PROJECTS

3D Neural Rendering Pipeline (MSc Project)

- Developed a **Python**-based pipeline to download, prepare, and pre-process large-scale 3D scene datasets for neural rendering experiments.
- Trained a neural network model using **PyTorch** on **GPU** instances to achieve photorealistic rendering, evaluating performance based on visual fidelity and computational efficiency.
- Implemented data augmentation techniques and hyperparameter tuning strategies to improve model convergence and output quality for novel scene generation.

Automated Reporting Service (HCL Technologies)

- Engineered a **Java Spring Boot** application that consumed data from **PostgreSQL** and generated daily operational reports, automating a process previously done manually.
- Developed a **Python** script to orchestrate report generation and distribution via email, reducing manual effort by approximately **15** hours per month.
- Integrated the service with **Jenkins** for scheduled execution and basic monitoring, ensuring reliable delivery of critical business insights.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)